Loss-versus-Rebalancing Cancellation in AMMs under Constant Volatility: A Reflected Skorokhod Equation Approach

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Abstract

We analyze fee income in a fee-only constant-product AMM arbitraged against a driftless GBM reference price. The main modeling step is a two-sided Skorokhod reflected SDE for the AMM–external log-mispricing: proportional fees define the no-arbitrage band, and the regulator records arbitrage-induced AMM price motion. In the stationary small-fee limit, the resulting invariant growth matches the classical constant-product LVR rate $\sigma^2/8$ to first order, leaving an $O(f^2\sigma^2)$ residual relative to continuous 50/50 rebalancing. Thus fees offset LVR to first order in this benchmark.

1 Introduction

Loss-versus-rebalancing quantifies the value extracted from passive AMM liquidity when stale AMM quotes are arbitraged against an external market. For a symmetric constant-product pool in the frictionless continuous-arbitrage limit, the classical loss rate relative to continuous 50/50 rebalancing is $\sigma^2/8$. This paper asks what happens to that leading term when proportional fees are included in the state dynamics.

The key point is that fees are not only a cashflow. They also create a no-arbitrage band for the AMM–external log-mispricing. In the fast-arbitrage regime, the natural continuous-time limit is therefore a reflected diffusion: the mispricing evolves inside the band, while arbitrage appears as the Skorokhod reflection needed to keep it there. The regulator has a direct AMM interpretation: its total variation is the cumulative arbitrage-induced motion of the AMM log-price.

Our contribution is to compute this benchmark explicitly. Under stationarity and constant external volatility, the reflection rate scales as $\sigma^2/(2\varepsilon)$. Combining this with the constant-product invariant-growth identity shows that, in the fee-only band $\varepsilon = -\log(1-f)$, fee-induced invariant growth equals $\sigma^2/8 + O(f^2\sigma^2)$. Hence the leading LVR term cancels, leaving only a second-order residual in the fee.

2 Preliminaries

We work with two assets: a numeraire asset and a risky asset. Throughout, all prices and wealths are measured in units of the numeraire. Let

- P_t^{ext} be the external market price of the risky asset;

- P_t^{amm} be the AMM marginal price of the risky asset;
- $S_t := \log P_t^{\text{ext}}$ be the external log-price;
- $A_t := \log P_t^{\text{amm}}$ be the AMM log-price;
- $Y_t := A_t - S_t$ be the AMM–external log-mispricing.

Thus $Y_t > 0$ means that the AMM price is above the external price, while $Y_t < 0$ means that it is below the external price.

Because arbitrage is not free, we do not impose $Y_t = 0$. Instead we assume a fixed no-arbitrage band

$$Y_t \in [-\varepsilon, \varepsilon] \qquad \text{for some } \varepsilon > 0. \tag{1}$$

The interpretation is that arbitrage is active only when the mispricing attempts to leave this band. This also assumes fixed costs of arbitrage are negligible.

2.1 Price model

The external market price is driftless geometric Brownian motion,

$$\frac{dP_t^{\text{ext}}}{P_t^{\text{ext}}} = \sigma\, dW_t, \qquad \sigma > 0. \tag{2}$$

Equivalently, the external log-price $S_t = \log P_t^{\text{ext}}$ satisfies

Assumption 1 (Driftless external price)**.**

$$dS_t = -\frac{\sigma^2}{2}\, dt + \sigma\, dW_t, \qquad \sigma > 0. \tag{3}$$

The relevant scale is the time needed for the log-price to move across a distance ε. Since a diffusive move over time t has size $\sigma\sqrt{t}$, the characteristic band-crossing time is

$$\tau_\varepsilon = \frac{\varepsilon^2}{\sigma^2}.$$

The log-price drift over this time scale is of size $(\sigma^2/2)\tau_\varepsilon = O(\varepsilon^2)$. We keep this drift in the model, but as we will see later, in the small-band regime it only contributes a lower-order correction to the quantities of interest.

2.2 When the SDE is a good approximation

In reality, arbitrage does not act continuously. Let $\tau > 0$ denote the reaction time (or block time), and let $t_i = i\tau$. Under Assumption 1, the external log-price increment over one step is

$$S_{t_{i+1}} - S_{t_i} = -\frac{\sigma^2}{2}\tau + \sigma\sqrt{\tau}\,\xi_{i+1}, \qquad \xi_{i+1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

Thus, before arbitrage acts, the mispricing evolves as

$$\widetilde{Y}_{t_{i+1}} = Y_{t_i} + \frac{\sigma^2}{2}\tau - \sigma\sqrt{\tau}\,\xi_{i+1}.$$

If $\widetilde{Y}_{t_{i+1}} \in [-\varepsilon, \varepsilon]$, no arbitrage trade occurs. If $\widetilde{Y}_{t_{i+1}} \notin [-\varepsilon, \varepsilon]$, the arbitrageur trades until the AMM price returns to the nearest boundary point.

In this paper, we use an SDE as the continuous-time approximation of this blockwise model. The approximation is accurate when a typical one-step move of the external log-price is small relative to the half-width of the no-arbitrage band:

$$\sigma\sqrt{\tau} \ll \varepsilon.$$

Equivalently, the reaction time must satisfy

$$\tau \ll \tau_\varepsilon.$$

In this regime, the overshoot beyond the band in a single step is small, so the discrete reset-to-boundary mechanism is well approximated by continuous reflection.

2.3 Standing assumptions

For the rest of the paper we work under the following assumptions.

Assumption 2 (Time-independent marginal law)**.** For every $t \geq 0$, Y_t has the same distribution as Y_0; equivalently,

$$Y_t \overset{d}{=} Y_0.$$

Assumption 3 (Fee-only small-fee regime)**.** The AMM charges a proportional fee $f \in (0, 1)$. We ignore gas costs, fixed costs, and other non-fee transaction costs, so the arbitrage band is determined only by the AMM fee and has half-width

$$\varepsilon = -\log(1 - f).$$

When asymptotic notation in f is used, it is understood in the limit $f \downarrow 0$.

Assumption 4 (Absence of organic microstructure)**.** Inside the band, the AMM price moves only because arbitrage pushes it back to the boundary. Equivalently, $dA_t = 0$ away from the reflection times. This isolates the fee mechanism and should be interpreted as a lower bound on AMM activity and profitability.

To keep notation light, we start the reflected system at time 0 under its stationary law and write all horizon-dependent quantities over $[0, t]$.

3 Skorokhod equation

The quantity of direct interest is the AMM log-price A_t, since fees are generated by the trades that move it. Under Assumption 4, the AMM does not move inside the band, so all motion comes from reflection at the boundary. From the definition of mispricing,

$$Y_t := A_t - S_t, \qquad \text{hence} \qquad dY_t = dA_t - dS_t,$$

the regulator admits a natural trading interpretation.

If the AMM becomes too cheap relative to the external market, then Y_t reaches the lower boundary $-\varepsilon$. Arbitrageurs buy the risky asset from the pool, which raises the AMM price; this upward correction is encoded by dK_t^+. If the AMM becomes too expensive, then Y_t reaches the

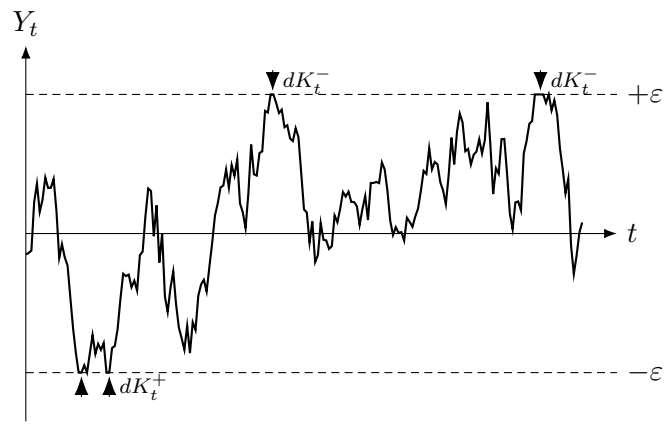

Figure 1: Skorokhod regulator: at the lower boundary, arbitrage raises the AMM price and contributes to K^+; at the upper boundary, arbitrage lowers the AMM price and contributes to K^-. The arrows indicate the direction of the regulator's action and are not meant to represent jumps.

upper boundary $+\varepsilon$. Arbitrageurs sell the risky asset into the pool, which lowers the AMM price; this downward correction is encoded by dK_t^-. Thus

$$dA_t = dK_t^+ - dK_t^-,$$

and, using (3),

$$dY_t = \frac{\sigma^2}{2}\,dt - \sigma\,dW_t + dK_t^+ - dK_t^-, \qquad Y_t \in [-\varepsilon, \varepsilon]. \tag{4}$$

This is a two-sided Skorokhod equation. The reflected diffusion problem was introduced by Skorokhod [1, 2] (see [3, 4] for English versions).

Here $K^\pm$ are nondecreasing, $K_0^\pm = 0$, and satisfy the Skorokhod support conditions

$$\int_0^t \mathbf{1}_{\{Y_u > -\varepsilon\}}\,dK_u^+ = 0, \qquad \int_0^t \mathbf{1}_{\{Y_u < \varepsilon\}}\,dK_u^- = 0. \tag{5}$$

These conditions say exactly that the corrections occur only when the mispricing is on the corresponding boundary. In the present model, all fee-generating activity is therefore concentrated at boundary interventions.

Define the total reflection

$$K_t := K_t^+ + K_t^-, \qquad \text{so } dK_t = dK_t^+ + dK_t^-. \tag{6}$$

This is the cumulative size of all arbitrage-induced corrections, without regard to sign.

Proposition 1 (AMM motion and total variation)**.** *For every $t \geq 0$,*

$$dA_t = dK_t^+ - dK_t^-, \qquad |dA_t| = dK_t, \qquad \mathrm{TV}(A; [0,t]) = K_t.$$

Proof. Combining $dY_t = dA_t - dS_t$ with (3) and (4) gives

$$dA_t = dY_t + dS_t = dK_t^+ - dK_t^-.$$

The Stieltjes measures dK^+ and dK^- are supported on the disjoint boundary sets $\{Y_t = -\varepsilon\}$ and $\{Y_t = \varepsilon\}$, hence are mutually singular. Therefore

$$|dA_t| = |dK_t^+ - dK_t^-| = dK_t^+ + dK_t^- = dK_t.$$

Integrating over $[0, t]$,

$$\mathrm{TV}(A; [0, t]) = \int_0^t |dA_u| = \int_0^t dK_u = K_t.$$

□

4 Expected accumulated reflection

Proposition 2 (Leading-order expected accumulated reflection)**.** *For every $t > 0$, in the small-band regime,*

$$\frac{1}{t}\mathbb{E}[K_t] = \frac{\sigma^2}{2\varepsilon} + O(\sigma^2\varepsilon). \tag{7}$$

Proof. Apply Itô's formula to Y_t^2 in (4). Since the regulator acts only at the boundary,

$$2Y_t\, dK_t^+ - 2Y_t\, dK_t^- = -2\varepsilon\, dK_t.$$

Hence

$$2\varepsilon K_t = Y_0^2 - Y_t^2 - 2\sigma \int_0^t Y_u\, dW_u + \sigma^2 t + \sigma^2 \int_0^t Y_u\, du. \tag{8}$$

Taking expectations, the martingale term vanishes and stationarity cancels the Y^2-terms. Therefore

$$2\varepsilon\mathbb{E}[K_t] = \sigma^2 t + \sigma^2 \int_0^t \mathbb{E}[Y_u]\, du.$$

By stationarity, $\mathbb{E}[Y_u] = \mathbb{E}[Y_0]$, so

$$2\varepsilon\mathbb{E}[K_t] = \sigma^2 t + \sigma^2 t\, \mathbb{E}[Y_0]. \tag{9}$$

It remains only to note that $\mathbb{E}[Y_0] = O(\varepsilon^2)$. Let $b := \sigma^2/2$. Taking expectations in (4) gives

$$\mathbb{E}[K_t^+ - K_t^-] = -bt. \tag{10}$$

Now apply Itô's formula to Y_t^3. Since $Y_t^2 = \varepsilon^2$ whenever the regulator acts, stationarity gives

$$0 = 3b\mathbb{E}[Y_0^2]t + 3\sigma^2\mathbb{E}[Y_0]t + 3\varepsilon^2\mathbb{E}[K_t^+ - K_t^-].$$

Using (10),

$$\mathbb{E}[Y_0] = \frac{1}{2}\big(\varepsilon^2 - \mathbb{E}[Y_0^2]\big) = O(\varepsilon^2),$$

because $|Y_0| \le \varepsilon$. Substituting this into (9) proves (7). □

5 Fee-induced invariant growth

For the constant-product pool, let x_t and y_t denote the reserves of the numeraire and risky assets, respectively, and define the invariant scale

$$D_t := 2\sqrt{x_t y_t}. \tag{11}$$

Since the AMM price is $P_t^{\text{amm}} = x_t/y_t = e^{A_t}$, the reserves can be written as

$$x_t = \frac{D_t}{2} e^{A_t/2}, \qquad y_t = \frac{D_t}{2} e^{-A_t/2}. \tag{12}$$

Proposition 3 (Invariant growth from AMM motion)**.** *For every $t \geq 0$,*

$$\log \frac{D_t}{D_0} = \frac{f}{4-2f} K_t. \tag{13}$$

Proof. By Proposition 1, A is continuous and of finite variation, with

$$\mathrm{TV}(A; [0,t]) = K_t.$$

Now consider one infinitesimal arbitrage trade, with input size u measured relative to the input reserve. In either direction,

$$\Delta \log D = \frac{1}{2}\Big[\log(1+u) - \log(1+(1-f)u)\Big],$$

while

$$|\Delta A| = \log(1+u) + \log(1+(1-f)u).$$

Using

$$\log(1+u) = u + O(u^2), \qquad \log(1+(1-f)u) = (1-f)u + O(u^2),$$

we obtain

$$\Delta \log D = \frac{f}{2} u + O(u^2), \qquad |\Delta A| = (2-f)u + O(u^2).$$

Hence

$$\Delta \log D = \frac{f}{4-2f} |\Delta A| + O\big((\Delta A)^2\big). \tag{14}$$

Now sum (14) over the individual arbitrage trades in a discrete approximation of the reflected evolution on $[0,t]$, and let the maximum trade size tend to zero. Because A is continuous and of finite variation,

$$\sum_i |\Delta A_i| \to \mathrm{TV}(A; [0,t]),$$

and

$$\sum_i |\Delta A_i|^2 \leq \Big(\sup_i |\Delta A_i|\Big) \sum_i |\Delta A_i| \to 0.$$

Therefore

$$\log \frac{D_t}{D_0} = \frac{f}{4-2f} \mathrm{TV}(A; [0,t]).$$

Substituting $\mathrm{TV}(A; [0,t]) = K_t$ gives (13). □

Corollary 1 (Small-fee invariant growth)**.** *In the fee-only small-fee regime, for every $t > 0$,*

$$\frac{1}{t}\mathbb{E}\left[\log \frac{D_t}{D_0}\right] = \frac{\sigma^2}{8} + O(f^2\sigma^2). \tag{15}$$

Proof. Taking expectations in (13) and using Proposition 2,

$$\frac{1}{t}\mathbb{E}\left[\log \frac{D_t}{D_0}\right] = \frac{f}{4-2f}\left(\frac{\sigma^2}{2\varepsilon} + O(\sigma^2\varepsilon)\right).$$

In the fee-only band, $\varepsilon = -\log(1-f) = f + f^2/2 + O(f^3)$, hence

$$\frac{f}{4-2f}\frac{1}{2\varepsilon} = \frac{1}{8} + O(f^2).$$

The remaining error is $O(f\sigma^2\varepsilon) = O(f^2\sigma^2)$, proving (15). □

6 LP performance relative to continuous rebalancing

Using (12), the value of the pool, measured in numeraire units and marked to the external price, is

$$V_t := x_t + P_t^{\text{ext}} y_t = D_t e^{S_t/2} \cosh\left(\frac{Y_t}{2}\right). \tag{16}$$

As benchmark, let $V_t^{50/50}$ denote the value of the self-financing portfolio kept continuously 50/50 by value in the numeraire token and the risky token at the external price.

Proposition 4 (Continuously rebalanced benchmark)**.** *For every $t \geq 0$,*

$$\log \frac{V_t^{50/50}}{V_0^{50/50}} = \frac{1}{2}(S_t - S_0) + \frac{\sigma^2}{8}t. \tag{17}$$

Proof. Let θ_t^n and θ_t^r denote the benchmark holdings of the numeraire and risky asset. By construction,

$$\theta_t^n = \frac{V_t^{50/50}}{2}, \qquad \theta_t^r = \frac{V_t^{50/50}}{2P_t^{\text{ext}}}.$$

Because the benchmark is self-financing and wealth is measured in numeraire units,

$$dV_t^{50/50} = \theta_t^r \, dP_t^{\text{ext}} = \frac{V_t^{50/50}}{2P_t^{\text{ext}}} \, dP_t^{\text{ext}}.$$

Using (2),

$$\frac{dV_t^{50/50}}{V_t^{50/50}} = \frac{\sigma}{2} \, dW_t.$$

Applying Itô's formula to $\log V_t^{50/50}$ gives

$$d \log V_t^{50/50} = \frac{\sigma}{2} \, dW_t - \frac{\sigma^2}{8} \, dt.$$

Since $dS_t = -(\sigma^2/2)dt + \sigma dW_t$, this is equivalently

$$d \log V_t^{50/50} = \frac{1}{2} \, dS_t + \frac{\sigma^2}{8} \, dt.$$

Integrating from 0 to t proves (17). □

For $t > 0$, define the expected excess log-return of the pool over this benchmark by

$$\Lambda_t := \frac{1}{t}\mathbb{E}\left[\log\frac{V_t}{V_0} - \log\frac{V_t^{50/50}}{V_0^{50/50}}\right]. \tag{18}$$

The main result of this section is that, in the fee-only band, Λ_t is second order in the fee.

Proposition 5 (Reduction to invariant growth)**.** *Under Assumption 2, for every $t > 0$,*

$$\Lambda_t = \frac{1}{t}\,\mathbb{E}\left[\log\frac{D_t}{D_0}\right] - \frac{\sigma^2}{8}. \tag{19}$$

Proof. From (16),

$$\log\frac{V_t}{V_0} = \log\frac{D_t}{D_0} + \frac{1}{2}(S_t - S_0) + \log\frac{\cosh(Y_t/2)}{\cosh(Y_0/2)}.$$

Subtract (17), take expectations, and divide by t. Since Y is stationary, Y_0 and Y_t have the same law, so

$$\mathbb{E}\left[\log\frac{\cosh(Y_t/2)}{\cosh(Y_0/2)}\right] = 0.$$

This gives (19). □

Theorem 1 (First-order cancellation of loss versus rebalancing)**.** *In the fee-only small-fee regime,*

$$\boxed{\Lambda_t = O(f^2\sigma^2).} \tag{20}$$

Thus fee-induced invariant growth cancels the classical constant-product loss-versus-rebalancing term to first order in the fee.

Proof. By Proposition 5 and Corollary 1,

$$\Lambda_t = \left(\frac{\sigma^2}{8} + O(f^2\sigma^2)\right) - \frac{\sigma^2}{8} = O(f^2\sigma^2).$$

□

7 Empirical implementation

Under the constant-volatility specialization, the leading-order model predictions are linear in the horizon length. Suppose external prices are observed on an evenly spaced grid

$$0 = t_0 < t_1 < \cdots < t_N = T, \qquad t_i - t_{i-1} = \Delta,$$

and write the observed external log-prices as

$$S_i := S_{t_i} = \log P^{\text{ext}}_{t_i}.$$

A natural estimator of the variance level is the realized-variance average

$$\widehat{\sigma}^2 := \frac{1}{N\Delta}\sum_{i=1}^{N}(S_i - S_{i-1})^2. \tag{21}$$

The small-fee predictions are

$$\frac{1}{T}\mathbb{E}[K_T] = \frac{\widehat{\sigma}^2}{2\varepsilon} + O(\widehat{\sigma}^2\varepsilon), \tag{22}$$

and, in the fee-only band $\varepsilon = -\log(1-f)$,

$$\frac{1}{T}\mathbb{E}\left[\log\frac{D_T}{D_0}\right] = \frac{\widehat{\sigma}^2}{8} + O(f^2\widehat{\sigma}^2), \qquad \Lambda_T = O(f^2\widehat{\sigma}^2). \tag{23}$$

8 Experiments

We numerically evaluate the leading-order prediction for expected accumulated reflection in (22). External log-price paths are generated using the discretization described above, with prescribed volatility σ and fixed grid spacing $\Delta = 1$ minute, or 60,000 milliseconds. The resulting paths are passed to a C++ simulator implementing the blockwise arbitrage mechanism: whenever the AMM–external log-mispricing leaves the no-arbitrage band $[-\varepsilon, \varepsilon]$, the AMM price is naturally reset to the nearest boundary via an arbitrage trade and the corresponding contribution to total reflection K_T is recorded.

For each pair (σ, f) on a parameter grid, with $\varepsilon = -\log(1-f)$, we compare the theoretical estimate $\sigma^2/(2\varepsilon)$ with the simulated value of K_T/T. The reported statistic is the Monte Carlo mean relative error

$$\mathbb{E}_{\mathrm{MC}}\left[\frac{\left|(K_T/T)_{\mathrm{sim}} - \sigma^2/(2\varepsilon)\right|}{\sigma^2/(2\varepsilon)}\right].$$

The same relative error can be nicely represented as a function of the dimensionless scale, with $\tau = \Delta$,

$$\sqrt{\frac{\tau}{\tau_\varepsilon}} = \frac{\sigma\sqrt{\tau}}{\varepsilon}, \qquad \tau_\varepsilon = \frac{\varepsilon^2}{\sigma^2},$$

which directly measures how large one-step moves are relative to the no-arbitrage band width. Figure 2 shows the resulting 3d plot. A broad intermediate region exhibits low relative error, indicating that the leading-order formula captures the simulated accumulated reflection rate over a wide range of volatility and fee-band parameters.

Figure 3 presents the same relative error against the collapsed scale $\sqrt{\tau/\tau_\varepsilon}$, making the scale-separation transition visually explicit.

The high-error regions have two distinct interpretations. When σ is large relative to ε, the typical one-step log-price displacement is large relative to the band width. This violates the scale separation condition $\sigma\sqrt{\Delta} \ll \varepsilon$ underlying the continuous-reflection approximation, so overshoots beyond the no-arbitrage band are no longer negligible. Conversely, when σ is small relative to ε, the characteristic crossing time $\tau_\varepsilon = \varepsilon^2/\sigma^2$ is large relative to the simulated horizon. Many trajectories then do not reach the boundary, producing little or no accumulated reflection. In this rare-boundary-hit regime, longer horizons or more Monte Carlo paths are needed to estimate K_T/T reliably.

9 Related works

Relation to loss-versus-rebalancing. The closest conceptual predecessor is the loss-versus-rebalancing framework of Milionis, Moallemi, Roughgarden, and Zhang [5]. Their continuous-time formula measures the predictable adverse-selection cost paid by passive liquidity providers when

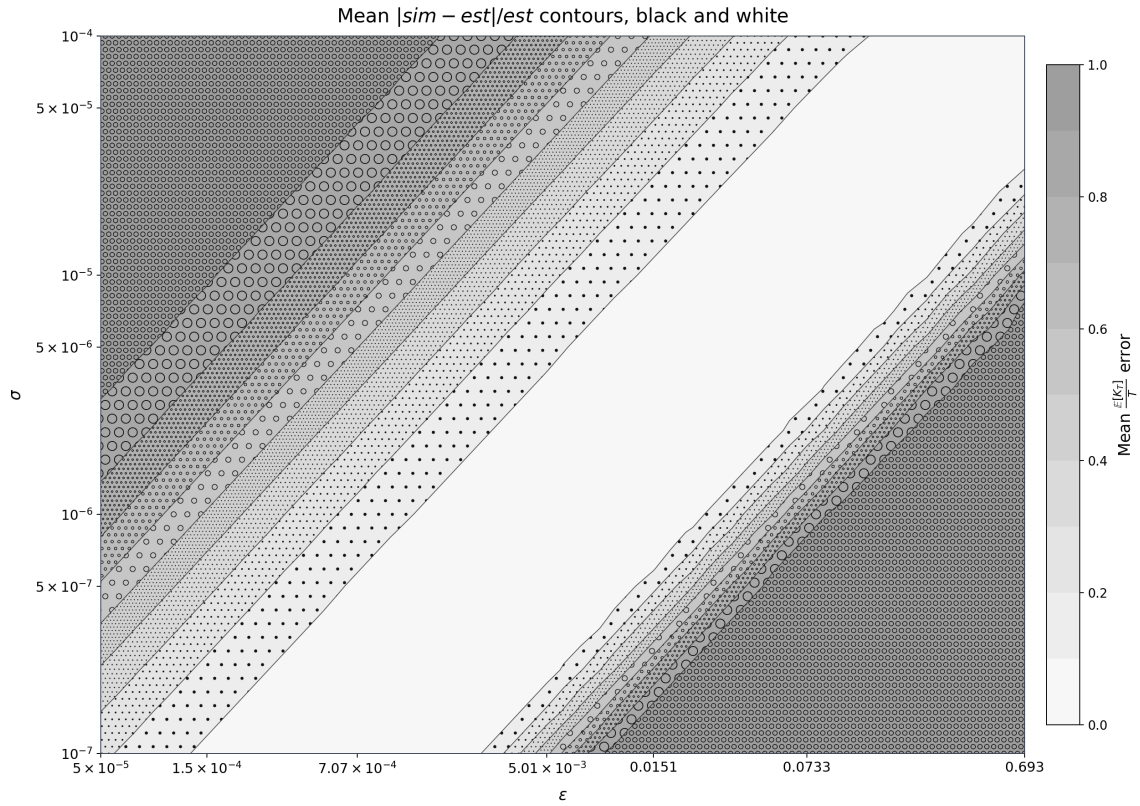

Figure 2: Mean relative error between the simulated reflection rate K_T/T and the leading-order estimate $\sigma^2/(2\varepsilon)$ over a grid of volatility and fee parameters, with $\varepsilon = -\log(1-f)$. The low-error region corresponds to the regime in which boundary hits are sufficiently frequent while one-step overshoots remain small.

stale AMM quotes are arbitraged against an external reference price; for a symmetric constant-product pool, this gives the familiar $\sigma^2/8$ rebalancing loss rate.

Our contribution is complementary. Rather than starting from the frictionless continuous-arbitrage limit, we put proportional fees into the state dynamics. The fees create the no-arbitrage band, and the Skorokhod regulator records the boundary trades that both move the AMM price and generate invariant growth. The cancellation theorem is therefore an LP-side benchmark for fee-induced growth relative to continuous rebalancing.

Relation to finite-block models. Trading fees and block production are studied more directly by Milionis, Moallemi, and Roughgarden [6], who analyze proportional fees with discrete Poisson block arrivals. Their arbitrage opportunities occur at block times and include overshoots beyond the fee threshold. Our model takes the scale-separated limit in which the blockwise reset mechanism is replaced by continuous reflection, appropriate when

$$\sigma\sqrt{\tau} \ll \varepsilon,$$

so that one-block external log-price moves are small relative to the band.

Nezlobin and Tassy [7] further refine finite-block analysis using deterministic and generalized block-time models. Their results emphasize the same dimensionless ratio appearing in our ex-

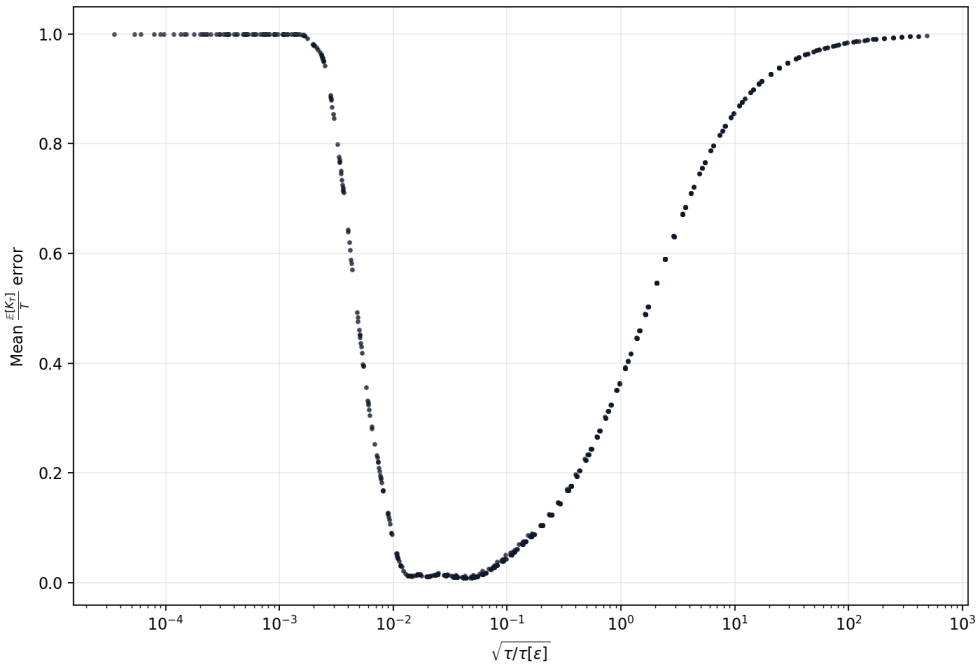

Figure 3: Mean relative error between simulation and the leading-order estimate, plotted against the scale ratio $\sqrt{\tau/\tau_\varepsilon}$. The low-error region is concentrated near small values of this ratio, consistent with the condition $\tau \ll \tau_\varepsilon$.

periments: $\sigma\sqrt{\tau}/\varepsilon$. When this ratio is not small, overshoots and rare boundary hits matter, and the reflected-diffusion approximation is no longer the right leading description. Consistently, our approximation

$$\frac{1}{t}\mathbb{E}[K_t] \approx \frac{\sigma^2}{2\varepsilon}$$

performs well in the scale-separated region and deteriorates outside it.

Dynamic fees. The reflected-diffusion viewpoint also gives a useful lens on dynamic fees. Recent work by Lebedeva, Umnov, Yanovich, Melnikov, and Ovchinnikov [8] studies adaptive asymmetric fees—including block-, deal-, and oracle-based rules—that reduce impermanent loss relative to fixed-fee baselines while preserving uninformed flow. In our model, such fees choose a time-varying asymmetric no-arbitrage band

$$[-\varepsilon_t^-, \varepsilon_t^+],$$

and the two Skorokhod regulators measure the resulting one-sided arbitrage pressure. Dynamic-fee design can therefore be viewed as adjusting the width and skew of this band using volatility, block time, inventory, and observed boundary local time.

Scope and outlook. Empirical work, including Fritsch and Canidio [9], studies arbitrage losses and fee income in actual AMM pools. Real LP performance depends on effects absent from the present model: organic order flow, gas and fixed costs, MEV competition, liquidity concentration, stochastic volatility, inventory changes, and the precise block-time mechanism. Our result should

therefore not be read as an empirical claim that AMM fees generally cover arbitrage losses. It identifies a clean local benchmark: in the stationary, driftless-GBM, constant-product, fee-only, fast-arbitrage limit, proportional fees offset the classical constant-product LVR term to first order.

The value of the Skorokhod formulation is that it separates three quantities that are often conflated: external volatility, the fee band, and invariant growth generated by boundary trades. This makes the model a tractable benchmark for extensions with finite block times, nonconstant volatility, non-fee transaction costs, organic order flow, dynamic fees, or more general AMM geometries.

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